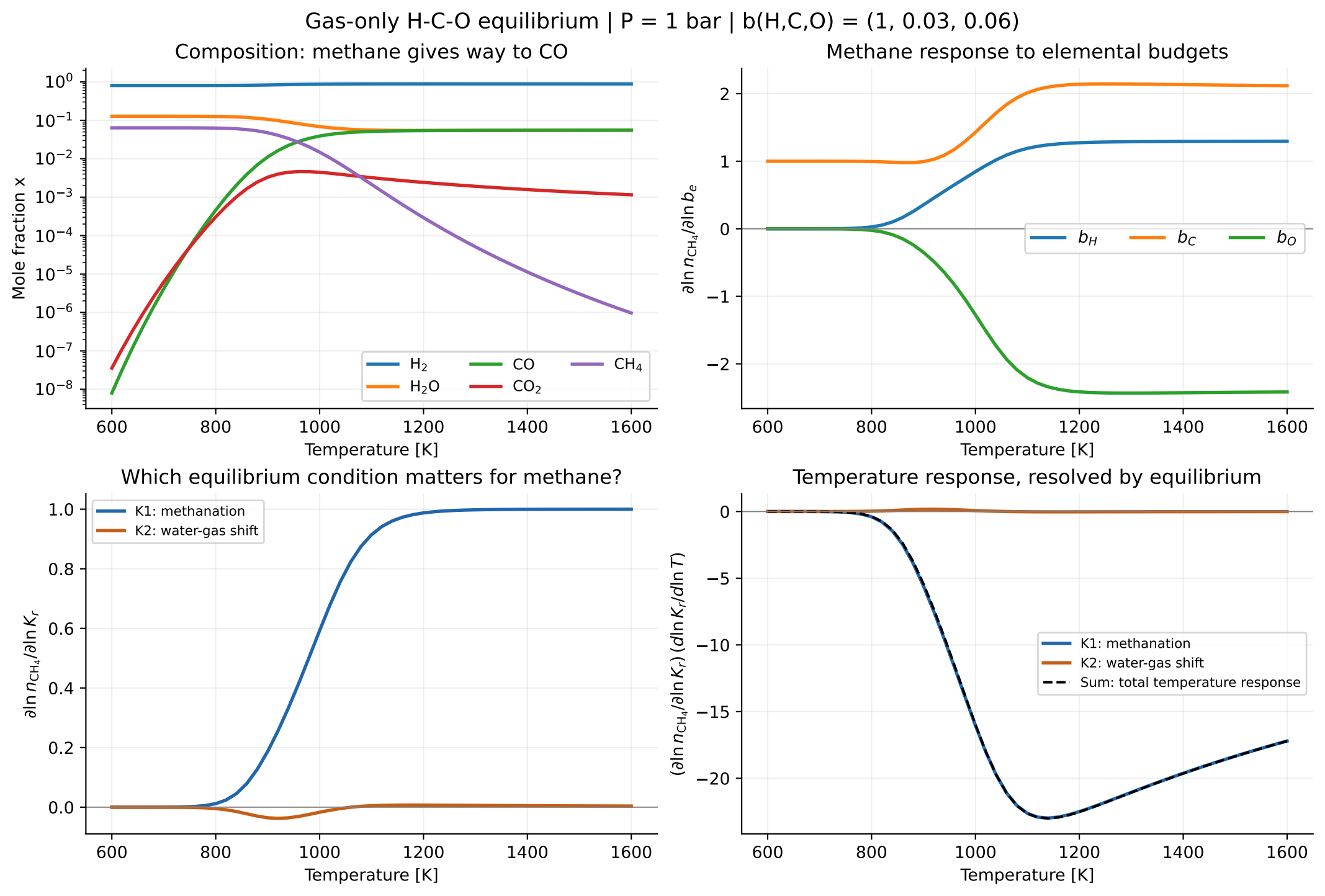
Gas-only H-C-O equilibrium | P = 1 bar | b(H,C,O) = (1, 0.03, 0.06)
Composition: methane gives way to CO
Mole fraction x
Temperature [K]
H2
H2O
CO
CO2
CH4
Methane response to elemental budgets
∂ln nCH4/∂ln be
bH
bC
bO
Which equilibrium condition matters for methane?
∂ln nCH4/∂ln Kr
K1: methanation
K2: water-gas shift
Temperature response, resolved by equilibrium
(∂ln nCH4/∂ln Kr) (dln Kr/dln T)
K1: methanation
K2: water-gas shift
Sum: total temperature response